



Results of TOR001: An open-label single patient study using targeted bacteriophage therapy for the treatment of chronic urinary tract infection

Jonathan D. Cook^{a,b,#}, Payton B. Hooey^{a,#}, Keiko C. Salazar^{c,d}, Justin R. Clark^{c,d}, Umesh Jain^a, Carlos Fernando^a, Hanjeong Harvey^e, Alexander Cervantes^d, Stephanie Omagari^d, Udi Blankstein^f, Andrew M. Kropinski^g, Alan R. Davidson^h, Karen L. Maxwell^h, Lori L. Burrows^e, Austen L. Terwilliger^{c,d}, Anthony W. Maresso^{c,d}, Greg J. German^{a,b,i,*}

^a Unity Health Toronto, St. Joseph's Health Centre, Toronto, Ontario, Canada

^b Department of Laboratory Medicine and Pathobiology, University of Toronto, Toronto, Ontario, Canada

^c Department of Molecular Virology and Microbiology, Baylor College of Medicine, Houston, TX, USA

^d Tailored Antibacterials and Innovative Laboratories for Phage (Φ) Research (TAILΦR), Baylor College of Medicine, Houston, TX, USA

^e Department of Biochemistry and Biomedical Sciences, McMaster University, Hamilton, Ontario, Canada

^f Department of Surgery, McMaster University, Hamilton, Ontario, Canada

^g Department of Pathobiology, University of Guelph, Ontario, Canada

^h Department of Biochemistry, University of Toronto, Toronto, Ontario, Canada

ⁱ Unity Health Toronto, Li Ka Shing Knowledge Institute, Toronto, Ontario, Canada

ARTICLE INFO

Article history:

Received 31 March 2025

Accepted 2 September 2025

Editor: Dr. Gina Suh

Keywords:

Bacteriophage

Phage therapy

Chronic urinary tract infection

Escherichia coli

ABSTRACT

Chronic urinary tract infections are persistent bacterial infections with the potential to drive antibiotic resistance. Like other persistent bacterial infections, intracellular bacterial reservoirs and biofilm formation hinder the clearance of pathogens despite long courses of antibiotic therapy. New strategies for treatment of these persistent infections are needed.

Here we describe the results of an open-label individual patient study using bacteriophage therapy to treat a chronic urinary tract infection in an immunocompetent 72-year-old woman without underlying urolithiasis or indwelling devices. We co-administered HP3.1, HP3, and ES19 bacteriophages with demonstrated *in vitro* activity via bladder instillation, orally, and as a topical formulation.

The primary outcome was safety and tolerability of the treatment. Adverse events were monitored through daily symptom logs and laboratory analysis of blood samples obtained throughout the study. We found that the treatment was safe and well tolerated with no serious adverse events reported. No adverse events were deemed related to the study material. The secondary outcomes were clinical and microbiological efficacy as monitored through daily symptom logs and standard urine culture, respectively. Two weeks after initial clinical improvement, her condition relapsed and culture was consistent with previous isolates. Treatment with ertapenem following bacteriophage therapy led to sustained clinical and microbiologic cure. Exploratory analysis through whole genome sequencing of pre- and post-treatment isolates identified mutations in genes associated with adhesion and evasion that may influence virulence and promote clearance.

These results inform expanded randomized clinical trials and support the growing literature that bacteriophage therapies are safe and effective.

© 2025 Published by Elsevier Ltd.

Abbreviations: AE, adverse event; AMR, antimicrobial resistance; CFU, colony forming unit; CDS, coding sequence; CLED, cystine-lactose-electrolyte deficient; Cr, creatinine; CRP, C-reactive protein; CT, computed tomography; DSMB, data safety monitoring board; Egfr, estimated glomerular filtration rate; ESBL, extended-spectrum beta-lactamase; ESR, erythrocyte sedimentation rate; IgE, immunoglobulin E; LAL, limulus amoebocyte lysate; LB, lysogeny broth; MALDI TOF, matrix-assisted laser desorption ionization time of flight; MOI, multiplicity of infection; phage, bacteriophage; Plt, platelets; PT, phage therapy; PVDF, polyvinylidene difluoride; RBC, red blood cells erythrocytes; REB, research ethics board; SAE, serious adverse event; SNP, single nucleotide polymorphisms; SPS, single patient study; UPEC, uropathogenic *Escherichia coli*; UTI, urinary tract infection; WBC, white blood cells (leukocytes).

* Corresponding author. Mailing address: Unity Health Toronto, St. Joseph's Health Centre, Toronto, Ontario, Canada.

E-mail address: Gregory.German@unityhealth.to (G.J. German).

These authors contributed equally.

1. Introduction

Advances in genetic sequencing and molecular biology have facilitated the characterization of bacteriophages (phage) for clinical trials and compassionate use [1]. Given the increasing burden of antimicrobial resistance (AMR), there is a renewed interest in the clinical application of phages as a novel anti-infective option, administered alone or in combination with traditional antibiotics [2,3].

Phages have been used experimentally to treat various infections including upper respiratory tract infections, urinary tract infections (UTIs), bacteremia, endocarditis, and osteomyelitis [4–7]. Successful treatment with phage therapy (PT) is attributed to their intrinsic ability to penetrate and destroy biofilms while reproducing at the source of infection [8]. Mounting evidence illustrates that PT and antibiotics have synergistic effects, including lower minimum inhibitory concentrations needed to treat drug-resistant organisms and re-sensitization of previously resistant isolates [9,10]. However, there is significant lead time for identification of suitable phages that can be used as therapy in an acute infection, necessitating careful selection of patient and infectious etiology when considering PT.

Urinary tract infections (UTIs) are one of the most common indications for antibiotic prescription in the community [11]. Overuse of antimicrobial medications for the treatment of UTIs outside of the hospital is as high as 50% and has a demonstrable impact on AMR [12–14]. UTIs recur in more than 1 in 4 women, requiring multiple courses of antibiotics [15]. Uropathogenic *Escherichia coli* (UPEC), one of the most common causes of UTI, can persist in intracellular reservoirs within the urinary bladder epithelium and via biofilm formation, leading to chronic infection [16,17]. These organisms can acquire plasmid-mediated resistance to antimicrobials from their environment and have a propensity to become resistant to multiple antibiotics, including extended-spectrum beta-lactams and carbapenems [18]. Treatment of chronic UTIs generally relies on the use of antibiotics to suppress symptomatic flares, rather than eradication.

Modern PT is an emerging field in infectious disease management, with the potential to address many challenges posed by both UTIs and AMR. Here, we describe the use of a targeted phage cocktail in the management of a chronic UTI caused by an extended-spectrum beta-lactamase (ESBL)-producing isolate of *E. coli*.

2. Materials and methods

2.1. Study design

This study was an open-label single patient study (SPS) of targeted phage administration for the treatment of a chronic UTI (NCT05537519). The primary outcomes addressed were the safety and tolerability of minimally invasive PT over a 90-day study period. The secondary outcomes were clinical and microbial responses over this same period. Exploratory outcomes included UTI recurrence over 6 months, induction of anti-phage humoral immune responses in the patient, and comparative genomics between bacterial isolates. Full protocol details are provided in Appendix A.

2.1.1. Regulatory approval

At the time of study design, Health Canada through their Biologics and Genetic Therapies Directorate provided guidance for the testing of investigational treatments or products in a specific patient via the SPS framework. Briefly, the framework requires that the specific patient meet the following criteria:

i) *Serious, life-threatening condition.*

ii) *Not eligible for, or exhausted all other treatment options.*

iii) *No other way to access treatment.*

An SPS is considered a clinical trial by Canadian regulatory bodies and can undergo a pre-submission teleconference with the study sponsor, followed by submission of a formal Clinical Trial Application package to Health Canada and registration with ClinicalTrials.gov if successful. The application is then reviewed by Health Canada with decisions made available within 30 days of submission.

The TOR001 SPS received regulatory approval via a “no objection letter” obtained from appropriate federal authorities within Health Canada and was reviewed and approved locally by the Unity Health Toronto Research Ethics Board (REB; 21-February-2023; REB 22-210). Informed consent was obtained from the participant in accordance with the International Conference on Harmonization - Good Clinical Practice.

2.1.2. Study population

The study population was a single participant with a severe, recurrent, and multi-drug resistant *E. coli* urinary tract infection. Inclusion criteria were females greater than or equal to 18 years of age who were previously diagnosed with a chronic and recurrent UTI. The study excluded participants with stage 5 chronic kidney disease, abnormal liver function tests, urinary stenting or chronic indwelling catheterization, known allergies to any component of the investigational medicinal product, current fever, and pregnancy. The participant was required to speak and understand English, be willing to follow the protocol, and could not be involved in another clinical trial.

2.1.3. Investigational biologic product

A single isolate of *E. coli* (Section 2.2) was obtained from standard culture of the participant’s urine during a symptomatic flare of their chronic UTI approximately 6 months prior to study day 1. All ESBL *E. coli* isolated during routine clinical care are frozen and stored at -80°C . Following informed consent and enrollment in the SPS, a phage cocktail with activity against this clinical isolate was manufactured by TAILOR service center (Baylor College of Medicine) and is herein referred to as UHT1. The UHT1 cocktail contains high-titre phage strains HP3 (1.84×10^{11} PFU/mL; Accession KY608967), HP3.1 (1.93×10^{11} PFU/mL; Accession OK275722), and ES19 (1.91×10^{11} PFU/mL; Accession MN508616) purified by CsCl gradient ultracentrifugation in Tris buffer with 100mM NaCl and 10mM MgCl_2 at pH 8.0 [19]. The UHT1 cocktail underwent endotoxin removal via Endotrap HD (LIONEX GmbH) column chromatography according to the manufacturer’s protocol for column processing and then filtered using a $0.22 \mu\text{m}$ polyvinylidene difluoride (PVDF) syringe filter. The resulting final product was aliquoted into 2mL glass vials with screw top caps and assayed for endotoxin and sterility.

Endotoxin testing was performed at the Center for Cell & Gene Therapy (Houston, Texas). A sample of the UHT1 phage cocktail was diluted 3 logs in endotoxin-free water prior to testing. A limulus amoebocyte lysate (LAL) assay was performed on this sample using a PTS201 Endosafe cartridge. Final endotoxin level of the UHT1 cocktail after purification was 485 EU/mL.

USP<71> and acid-fast bacteria (AFB) sterility testing was performed by Nova Biologicals (Conroe, Texas). For USP<71> testing, samples were incubated for no less than 14 days in both Thioglycollate broth ($32.5^{\circ}\text{C} \pm 2.5^{\circ}\text{C}$) and Tryptic Soy broth ($22.5^{\circ}\text{C} \pm 2.5^{\circ}\text{C}$). The media was examined daily for macroscopic evidence of microbial growth during the incubation period. For AFB testing, samples were incubated at 30° to 35° for a total of 8 weeks in Lowenstein Jensen Media.

Sequencing of each phage was performed by the manufacturer to optimize safety and to predict genus, attachment sites, and

Table 1
TOR001 schedule of events.

	Study day										
	0	1	2	3	4	5	7 [†]	14 [^]	30 [^]	60 [^]	90 [^]
Oral pantoprazole administration		X									
Antibiotic administration						X [†]	X [†]	X [†]	X [†]	X [†]	X [†]
UHT1 bladder instillation		X					X [*]				
UHT1 oral		X					X [*]				
UHT1 topical application			X	X	X						
Blood sampling		X	X		X		X	X	X	X	X
Vaginal swab		X	X	X	X	X	X	X	X	X	X
Urine sampling	X	X	X	X	X	X	X	X	X	X	X

* Dependent on sample results provided on Day 5.

† Dependent on patient symptoms and biological samples.

[^] ± 2 days for visits on days 14, 30, 60 & 90.

phage lifestyle. Coding sequences (CDS) were analyzed for genes encoding virulence factors, integrases, and antibiotic resistance.

2.1.4. Study interventions

The participant was assessed for the continued presence of multi-drug resistant *E. coli* via standard urine culture following cessation of prophylactic antibiotics and recrudescence of lower urinary tract symptoms. The UHT1 cocktail was subsequently administered via three separate routes including oral, topical, and via bladder instillation. On day 1 of the study, an oral solution of UHT1 at a dose of 4.3×10^{11} PFU in 100 mL 6% sodium bicarbonate was administered following a single dose of pantoprazole 40 mg PO. On the same visit day, the participant underwent bladder instillation of UHT1 at a dose of 4.3×10^{11} PFU in 100 mL normal saline. The UHT1 cocktail was retained for 1 hour with position changes every 15 min. On subsequent days 2 through 4, a topical formulation of UHT1 was applied directly to the urethral opening with a sponge applicator for 30 min at a dose of 2.6×10^{10} PFU. Vaginal swabs, urine, and blood samples were collected for analysis according to the Schedule of Events (Table 1). Stool testing for ESBL *E. coli* carriage was not possible due to availability of testing at the clinical microbiology laboratory so samples were not obtained. This minor protocol deviation affected an exploratory endpoint and was brought to the attention of the Sponsor-Investigator and discussed with the local REB. A technical memo was issued but no amendment to the Study Protocol was deemed necessary by the REB.

2.1.5. Safety monitoring

Blood samples were collected on study day 1, prior to the administration of the UHT1 cocktail, and subsequently on study days 2, 4, 8, 15, 19, 26, 29, 58, and 86. Complete blood counts, inflammatory markers, liver profile, kidney function, and quantitative immunoglobulin E (IgE) were assessed according to the study protocol.

A daily symptom log was maintained by the participant throughout the entirety of the study and for 90 days following the date of the last intervention. Monitoring of adverse events (AEs) was the sole responsibility of the Investigator and was to be recorded in the patient's medical record and on the AE Case Record Form. AEs were categorised according to the National Cancer Institute Common Terminology Criteria for Adverse Events version 4 as grade 1 to 5. A data safety monitoring board (DSMB) comprised of physician experts received notifications weekly during the active components of the study. After the active treatment phase, notifications continued on a quarterly basis.

Any serious adverse events (SAEs) were to be documented and reported to the DSMB and the Research Ethics Board within 24 h of occurrence. SAEs that were unexpected and deemed to be related to the study treatment were to be reported to the regulatory authorities as per International Conference on Harmonization Guide-

lines and through the Health Canada Adverse Drug Reaction Expedited Reporting Summary Form as further described in Appendix A. Additionally, blood work results greater than 3x the upper limit of normal or fever (defined as body temperature greater than or equal to 38.0°C) required the immediate suspension of the study.

2.2. Bacterial isolate characterization

2.2.1. Urine culture procedures

Mid-stream urine was collected from the participant for surveillance at screening and on days 1, 2, 3, 4, 5, 8, 9, 15, 29, 58, and 86 of the study. Unscheduled urine cultures for the purpose of clinical care were obtained on study days 19 and 26. Urinalysis and microscopy were performed on a subset of these samples (Table 2). Urine cultures were performed and interpreted according to local clinical microbiology laboratory procedures. Briefly, 10 µL of urine was plated on cystine-lactose-electrolyte deficient (CLED) agar plates and incubated at 37°C for 18 h. Colonies were enumerated semi-quantitatively and isolates were identified to the species level using matrix-assisted laser desorption ionization time of flight (MALDI TOF) mass spectrometry.

2.2.2. Antibiotic susceptibility testing

Antibiotic susceptibility testing was performed on the VITEK®2 platform (bioMérieux) and the Ast-N391 (bioMérieux) antimicrobial susceptibility testing cards. ESBL production was screened via phenotypic testing according to local clinical microbiology laboratory procedures.

2.2.3. Bacterial whole genome sequencing

Urine *E. coli* isolates from the initial clinical sample (pre-treatment) and post-treatment cultures (D15) were subjected to whole genome sequencing. Briefly, commercial DNA extraction was performed from a single colony of each isolate (Génome Québec, QC). Libraries were prepared using Illumina DNA Prep (Illumina, San Diego, CA) with IDT for Illumina Unique Dual (UD) Indexes (Illumina, San Diego, CA) according to the manufacturer's instructions. Individual libraries were quantified using the Qubit 4 Fluorometer with the Qubit 1X dsDNA High Sensitivity (HS) kit and normalized to equimolar concentrations. Pooled libraries were analyzed on the TapeStation 4150 using D1000 ScreenTape (Agilent Technologies, Santa Clara, CA) and sequenced on the MiniSeq platform using a 300-cycle HighOutput Reagent kit (Illumina, San Diego, CA) in paired-end mode (2 × 149 bp).

2.2.4. Whole genome sequencing analysis

Reads obtained from short-read sequencing were assembled *de novo* using Geneious Assembler with medium-low sensitivity settings in Geneious Prime (version 2025.0.2) after reads were

Table 2
Results of blood work and urine analyses throughout the TOR001 study.

		Study Day												
		1	2	3	4	5	8	9	15	19	26	29	58	86
Complete Blood Counts	Hemoglobin (g/L)	139	134	–	130	–	130	–	132	131	126	126	131	126
	RBC (10 ¹² /L)	4.5	4.3	–	4.3	–	4.2	–	4.3	4.3	4.1	4.1	4.2	4.2
	WBC (10 ⁹ /L)	7.8	8.8	–	8.1	–	8.3	–	9.5	6.9	9.5	6.5	8.4	6.0
	Plt (10 ⁹ /L)	331	341	–	331	–	341	–	323	368	352	350	311	311
Inflammatory Markers	CRP (mg/L)	2.5	2.3	–	2	–	3.7	–	58.3	8.1	16.3	4.1	–	–
	ESR (mm/Hr)	–	18	–	21	–	21	–	34	–	–	19	14	15
Liver Profile	AST (U/L)	22	16	–	16	–	23	–	14	23	22	16	–	–
	GGT (U/L)	18	15	–	14	–	15	–	14	–	–	11	–	–
Kidney Function	Cr (μmol/L)	109	96	–	94	–	92	–	87	86	82	83	–	–
	eGFR (mL/min/1.73m ²)	47	55	–	56	–	58	–	62	63	66	65	–	–
Immunology	IgE (kIU/L)	–	26.2	–	25.8	–	27.4	–	30.3	33.5	28.3	27.7	25.8	–
Urinalysis	Colour & Appearance	–	Yellow; cloudy	Yellow; cloudy	–	Yellow; clear	Yellow; clear	Yellow	Yellow	Yellow	Yellow	–	–	–
	Specific Gravity	–	1.010	1.011	–	1.008	1.011	1.010	1.008	≤1.010	≤1.005	–	–	–
	pH	–	6.5	6.0	–	6.0	5.5	6.0	5.5	6.0	6.0	–	–	–
	Leukocytes – Macroscopic (cells/uL)	–	500	500	–	125	70	500	125	125	15	–	–	–
	Nitrites (binary)	–	NEG	NEG	–	NEG	NEG	NEG	NEG	NEG	NEG	–	–	–
	Protein (g/L)	–	TRACE	NEG	–	NEG	NEG	0.3	NEG	NEG	NEG	–	–	–
	Glucose (mmol/L)	–	NEG	NEG	–	NEG	NEG	NEG	NEG	NEG	NEG	–	–	–
	Ketones (mmol/L)	–	NEG	NEG	–	NEG	NEG	NEG	NEG	NEG	NEG	–	–	–
	Blood (cells/uL)	–	MOD	TRACE	–	NEG	NEG	25	NEG	NEG	NEG	–	–	–
	WBC (cells/hpf)	–	>50	>50	–	21-30	31-50	≥50	31-50	10-20	0-2	–	–	–
	RBC (cells/hpf)	–	3-5	0-2	–	0-2	0-2	3-5	0-2	0-2	0	–	–	–
	Bacteria	–	3+	0	–	0	0	1+	0	Occ	0	–	–	–

RBC, erythrocytes; WBC, leukocytes; NEG, negative; MOD, moderate; TRACE, below quantification; Occ, occasional; Plt, platelets; CRP, C-reactive protein; ESR, erythrocyte sedimentation rate; AST, aspartate aminotransferase; GGT, gamma-glutamyl transferase; Cr, creatinine; eGFR, estimated glomerular filtration rate; IgE, immunoglobulin E.

trimmed to a quality score of 20 and reads shorter than 30 bps were filtered out using BBduk (version 38.84; <https://github.com/BioInfoTools/BBMap>). Contigs under 1,000 base pairs were discarded to produce the final assemblies. Assemblies were then annotated using the RASTtk pipeline [20–22] and MLST-typed using the PubMLST database [23]. The *fimH* subtype for the isolates was determined by aligning the *fimH* alleles from FimTyper [24] to the *fimH* sequence for both the pre- and post-treatment isolates using MAFFT (v7.490) [25].

Snippy (<https://github.com/tseemann/snippy>) with default parameters was then used to call variants with the pre-phage treatment isolate *de novo* assembly used as a reference and the post-treatment isolate trimmed and filtered reads used for variant calling. Verification was performed via direct genome alignment and comparison. *De novo* assemblies were aligned using Mauve Genome Aligner (version 2015-02-26) [26] with the Mauve Contig Mover algorithm with default settings. Dissimilarities between the two sequences were extracted using Geneious Variant Finder with default settings and the pre-treatment isolate was used as the reference sequence.

The genome of the initial clinical isolate was assembled using long-read sequencing, which resolved the complete chromosome and two plasmids. Pre- and post-treatment reads from short-read sequencing were mapped to the complete ancestral chromosome and plasmids, and the resulting consensus sequences for the chromosome and both plasmids were aligned using Mauve Genome Aligner to further verify the Snippy results.

2.3. Phage characterization

2.3.1. Phage susceptibility testing

Bacterial isolate susceptibility to the UHT1 phage cocktail was assessed using a standard quantitative killing assay. Briefly, pre- and post-treatment isolates were grown to log phase and 10^6 colony forming units (CFUs) were seeded into 100 μ L lysogeny broth (LB) media in a 96-well plate. Individual phage and the UHT1 cocktail were separately added to the wells at varying multiplicity of infection (MOI) and incubated at 37°C for 4 h with shaking. Serial dilutions of the cultures and the starting inoculum were then plated on LB agar plates. The plates were incubated at 37°C overnight and colonies counted. Results are reported as the average $\pm$ standard deviation of 3 technical replicates. Comparison of mean differences between the pre- and post-treatment isolates was performed and statistical significance was determined using Student's t test or Mann-Whitney U test, as appropriate. Normality for all data sets was determined using the Shapiro-Wilk Test. Statistics were calculated using GraphPad Prism 10.4.0.

2.3.2. Comparative phage titer assay

Following treatment, UHT1 cocktail and the individual phages that comprise it were serially diluted and titered on a lawn of either pre- or post-treatment isolates in triplicate. Comparison of mean differences between the pre- and post-treatment isolates was performed and statistical significance was determined using Student's t test. Normality for all data sets was determined using the Shapiro-Wilk Test. Statistics were calculated using GraphPad Prism 10.4.0.

2.3.3. Serum neutralization assays

To test for the generation of neutralizing antibodies, 100 μ L fractions of serum obtained throughout the study were incubated with 1 μ L of UHT1 cocktail in 96-well microtiter plates for 30 min at 37°C. The phage-serum mixtures were then serially diluted in phage buffer and titered on the pre-treatment bacterial isolate. This process was performed three times, each time with three replicates per serum sample [27].

3. Results

3.1. SPS recruitment and participant characteristics

The participant recruited to the TOR001 SPS was a 72-year-old Caucasian female with a chronic UTI caused by ESBL *E. coli*. The participant had a past medical history significant for diverticulitis, urinary retention, and recurrent UTIs. The latter ultimately resulted in unilateral ureteral reconstruction in 2017 and a right nephrectomy in 2019 secondary to inflammatory damage wrought by persistent infection. Although her initial isolates were susceptible to first line antimicrobial therapies, the participant became chronically infected post-nephrectomy with an ESBL *E. coli*. She remained on suppressive antibiotic therapy and was cycled through active oral antimicrobial agents to minimize the generation of additional resistance. Multiple breakthrough infections required treatment with intravenous carbapenems. Over the ensuing 6 years, the patient did not clear the ESBL *E. coli* despite several extended courses of antibiotics including multiple extended courses of ertapenem. Of note, the patient did not have evidence of ESBL *E. coli* carriage in her stool. After culture confirmation of persistent ESBL *E. coli* infection, optimization of genitourinary health with topical estrogen, and negative workup for underlying causes of her chronic infections including cystoscopy and computed tomography (CT), the participant was recruited into the study. Her symptoms at screening included a burning sensation at the urethral opening, frequency, and nocturia, but were considered manageable.

The clinical isolate of ESBL *E. coli* identified in routine clinical samples of urine from the participant was sent to TAILOR Labs (Baylor College of Medicine, Texas) for screening against a subset of 23 phages from their library. Phage susceptibility testing identified four phages with lytic activity against the ESBL-producing *E. coli* isolate, of which three (HP3, HP3.1 and ES19) were subsequently used in producing the UHT1 phage cocktail (Fig. 1A). All three of these phages are myoviridae/myovirus by sequencing analysis. There is evidence that HP3 facilitates attachment via OmpA and the inner core of LPS, while HP3.1 (a variant of HP3) is able to circumvent resistance gained by inner core mutations in LPS synthesis [28]. Phage ES19 does not have a verified receptor.

At baseline and prior to administration of the UHT1 phage cocktail, the participant had normal complete blood counts, normal inflammatory markers, and normal liver function tests. Creatinine was mildly elevated corresponding to an estimated glomerular filtration rate (eGFR) of 47 mL/min/1.73m². Results of urinalysis and blood work throughout the study are presented in Table 2. The presence of ESBL *E. coli* was confirmed via urine culture on study day 1.

3.2. Safety and tolerability of the UHT phage cocktail

The primary endpoint of the TOR001 SPS was assessment of safety and tolerability of PT when administered orally, once via bladder instillation, and as a topical formulation. Safety was defined by the absence of SAEs attributable to study material occurring from study day 1 until study day 90. No SAEs occurred during the study. Weekly reports were issued to the DSMB for all AEs for the first 4 weeks and then quarterly for the remainder of the study.

The daily symptom diary was used to identify expected and unexpected AEs throughout the entirety of the SPS. The most common AE was rash. On study days 10, 11, 19, 20 and 21, the participant developed a bilateral pruritic rash on the dorsal aspect of their hands. This was managed with topical clotrimazole/betamethasone cream. Additionally, on study days 44, 45, 46, 50, 61, 65, 66 and 67 the participant experienced a painful genital rash which resolved without medication. The rash was inconsistent with herpes simplex virus infection. Consistent Nugent scores

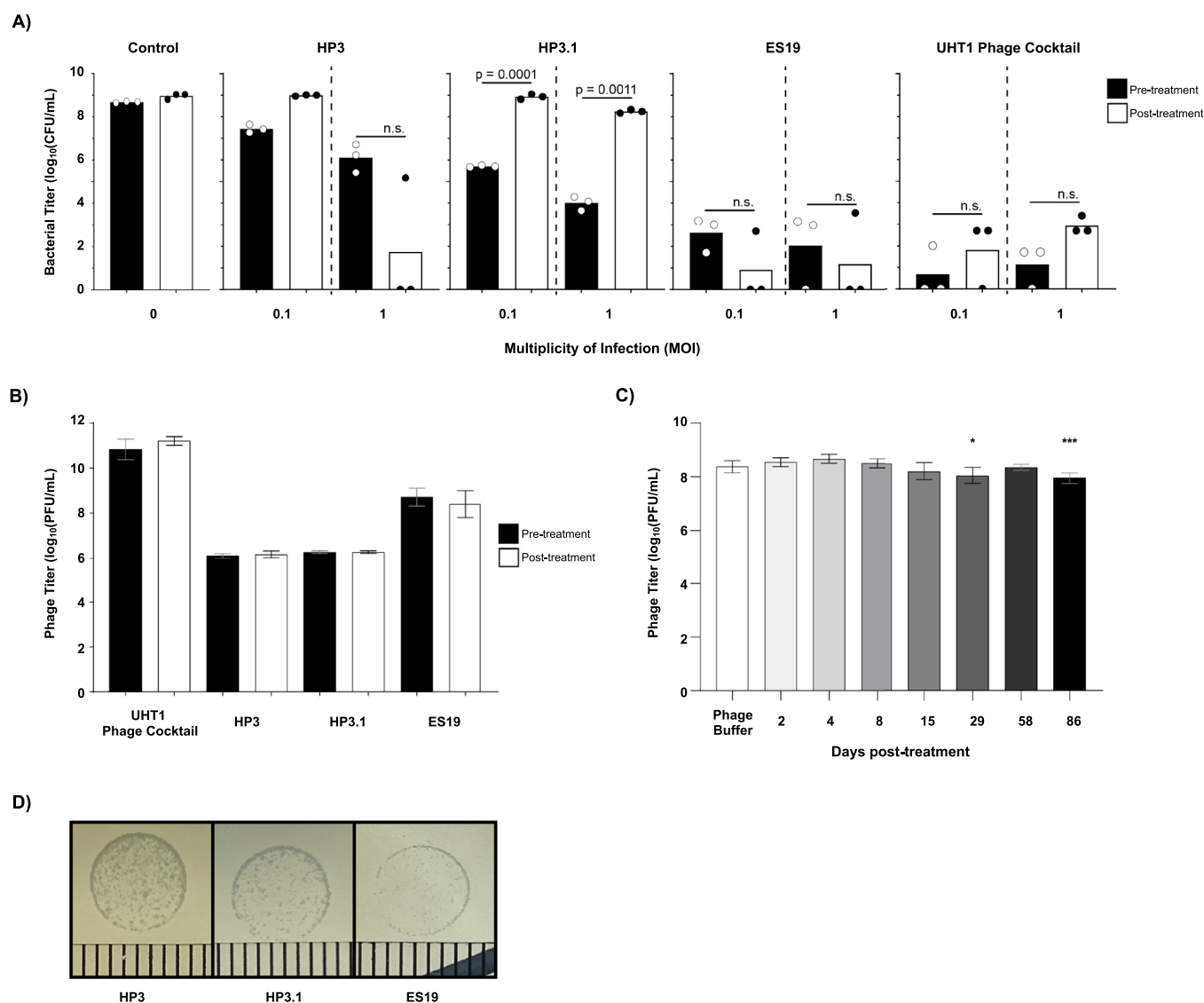


Fig. 1. *In vitro* characterization of the UHT1 cocktail. (A) Phage susceptibility testing shown as bacterial titer after challenge at a multiplicity of infection (MOI) of 0.1 and 1 by the UHT1 cocktail as well as individual phage components. MOIs were calculated via phage titer on DH5 α . Results are reported as the average $\pm$ standard deviation of 3 technical replicates. (B) Titer of the individual phages and cocktail on pre- (black bars) and post-treatment (white bars) isolates. Bars indicate the average ($\log_{10}$) titer between three biological replicates. Error bars represent standard deviation. There was no significant difference in phage susceptibility between the two isolates. The two isolates were compared to each other by Student's t-test. (C) Neutralization assay demonstrating the titer of UHT1 cocktail on the pre-treatment isolate after incubation with serum collected from the patient at multiple time points following phage therapy. Bars indicate the average ($\log_{10}$) titer of the cocktail between three biological replicates (each with three technical replicates). Error bars represent standard deviation. Statistics were calculated using GraphPad Prism 10.4.0. Each day was compared to the phage buffer control using Student's t-test. *, **, and *** represent p values of < 0.05, < 0.01, and < 0.001, respectively. (D) Representative plaque morphology of HP3, HP3.1, and ES19 on the pre-treatment isolate. Ruled lines = 1 mm.

of 0-3 throughout the entirety of the study demonstrate the maintenance of normal vaginal flora following topical application.

On study day 14 the participant presented to care describing abdominal pain characteristic of their prior diverticulitis. This was managed expectantly as an outpatient with a full-fluid diet for 14 days. A CT scan of the abdomen at that time identified mild inflammatory changes at the distal sigmoid colon and supported a diagnosis of uncomplicated diverticulitis. Daily symptom diaries were reviewed for a total of 6 months with monthly clinical follow up after day 86. The participant reported sustained symptom resolution and there were no expected or unexpected AEs reported after study day 67.

There were two instances of laboratory AEs, which are defined as a clinically significant change from baseline bloodwork. Elevations in inflammatory markers were noted on study day 15 and study day 26 corresponding with her diagnosis of diverticulitis. In

both instances, the markers returned to within normal limits at 96 hours and 72 hours, respectively. Of note, the participant had an improvement from pre-phage baseline in their eGFR to >60 mL/min/1.73m² by study day 19.

3.3. Clinical and microbiological efficacy of the UHT phage cocktail

Mid-stream urine was collected for culture analysis at screening and throughout the study period. After study day 1, urine samples had oscillating levels of *E. coli* and mixed growth of other bacteria until clearance on study day 9. Urinalysis showed a rapid decrease in leukocytes and bacteriuria by microscopy. Clinically significant outgrowth by ESBL-producing *E. coli* was seen as part of surveillance cultures on study days 2, 8, and 15, however, the participant reported complete resolution of symptoms within 48 h of the first treatment. Dysuria and progressive pain at the urethral

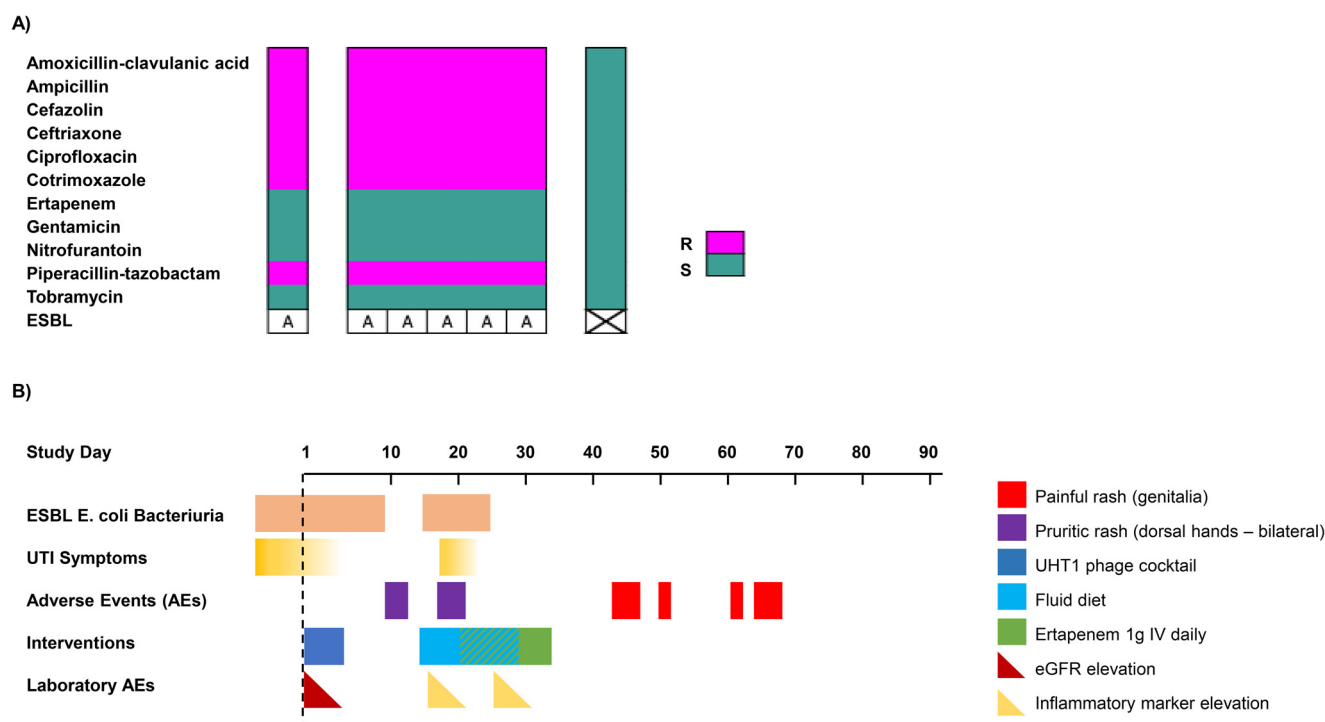


Fig. 2. Timeline of significant events throughout the TOR001 study period. A) Graphic representation of phenotypic data obtained from urine culture results. R (resistant); S (susceptible); A (ESBL Ambler Class A). B) Timeline summarizing data from daily symptom logs, results of blood and urine sampling, and from the clinical history of the participant.

opening on study days 19 through 21 prompted additional urine cultures, which again grew out ESBL *E. coli*. The patient remained afebrile and was provided a 14-day course of ertapenem as an outpatient for the treatment of UTI with rapid symptomatic response. Following treatment with ertapenem, the participant continued to be free of lower urinary tract symptoms; surveillance urine cultures were clear of ESBL-producing *E. coli* for the remainder of the study. On study day 86, surveillance cultures identified asymptomatic bacteriuria with non-significant outgrowth of pan-sensitive *E. coli*. Complete bacterial culture results and antibiotic susceptibilities are presented in Table 3. A timeline of bacterial culture results, recorded AEs, interventions, and symptoms are presented in Fig. 2A and Fig. 2B.

The susceptibility of pre- and post-treatment isolates to the HP3, HP3.1, ES19, and complete UHT1 cocktail was compared. Although there were no differences in bacterial titer between isolates when tested against the UHT1 cocktail in a standard killing assay, there were statistically significant increases in CFU/mL observed when HP3.1 was assayed against the post-treatment isolate alone (Fig. 1A). Additionally, no significant difference in phage titer was observed between the two isolates (Fig. 1B). These results support the strategy of using multiple active phages formulated as a cocktail to ameliorate the development of rapid phage resistance.

Serum from the participant obtained on study days 2, 4, 8, 15, 29, 58, and 86 was assayed for the ability to neutralize phage activity against the original pre-treatment isolate (Fig. 1C). There was no significant neutralization activity observed at any point in the study. Representative plaque morphology is presented in Fig. 1D.

3.4. Comparative genomics between UHT1 pre- and post-treatment *E. coli* isolates

The pre- and post-treatment isolates were sequenced to 108.7x and 33.7x coverage depth, respectively. For contigs >1000 base pairs in size, the *de novo* assemblies contained 54 contigs for both

the pre- and post-treatment isolate. Multilocus sequence typing assignment identified the isolates as ST131 *E. coli*.

Reads from pre- and post-phage treatment were mapped to *E. coli* EC958 (accession: HG941718.1), a high-quality ST131 reference genome, resulting in high identity (99.7%) but relatively low coverage (91.5%). Manual inspection of the *fimH* mapping confirmed there were multiple genetic differences between the clinical isolates and EC958. Further alignment of *fimH* alleles showed that the clinical isolates belonged to the *fimH41* subtype of ST131. A high-quality *fimH41* ST131 genome, denoted as strain WP5-S18-ESBL-09 (accession: AP022171) [29], was identified and used for mapping using Geneious Assembler in Geneious Prime 2025.0.2 (<https://www.geneious.com/>).

De novo assemblies were aligned and dissimilarities between the two sequences were extracted. There were 491 insertions, deletions, and single nucleotide polymorphisms (SNPs) identified between the pre- and post-treatment isolates. Of these, there were 84 differences that resulted in changes in 19 proteins. Several notable mutations are described in Table 4 and visualized by circus plot in Fig. 3. Notable mutations were verified by aligning post-treatment reads to the pre-treatment *de novo* assembly using Snippy and by mapping short-reads to the genome from the initial clinical genome that was resolved using long-read sequencing. All differences in the genome alignment between the pre- and post-treatment isolates are presented in Appendix B.

There were two plasmids identified that contained genes associated with ESBL *E. coli* phenotype (*blaCTX-M* and *blaEC*, respectively). Short-reads were also mapped to the plasmids found in the initial clinical isolate and the resulting consensus sequences for pre- and post-treatment were aligned, but showed no significant changes.

4. Discussion

Here we report the results of using a targeted phage cocktail in the treatment of a person living with chronic UTI caused by ESBL

Table 3

Complete urine culture results throughout the TOR001 study.

Interpretation	Study Day																						
	Screen	1		2	3			4			5			8	9	15	19	26	29	58	86		
					NSG			Probable contamination			Probable contamination				NSG			NG			NG	NSG	
Organism	E coli	E coli	GBS	E coli (>100)	E coli	GBS	E faecalis	E coli	GBS	E faecalis	E coli	GBS	S haemolyticus	E coli		E coli	E coli					E coli	E faecalis
Growth (CFU)	10-100	10-100	10-100	>100	1-10	10-100	1-10	>100	10-100	10-100	10-100	1-10	1-10	10-100	10-100	10-100	>100					10-100	10-100
Amoxicillin-clavulanic acid	R	R	–	R	–	–	–	–	–	–	–	–	–	–	R	–	R	R	–	–	–	S	–
Ampicillin	R	R	–	R	–	–	–	–	–	–	–	–	–	–	R	–	R	R	–	–	–	S	S
Cefazolin	R	R	–	R	–	–	–	–	–	–	–	–	–	–	R	–	R	R	–	–	–	S	–
Ceftriaxone	R	R	–	R	–	–	–	–	–	–	–	–	–	–	R	–	R	R	–	–	–	S	–
Ciprofloxacin	R	R	–	R	–	–	–	–	–	–	–	–	–	–	R	–	R	R	–	–	–	S	–
Cotrimoxazole	R	R	–	R	–	–	–	–	–	–	–	–	–	–	R	–	R	R	–	–	–	S	–
Ertapenem	S	S	–	S	–	–	–	–	–	–	–	–	–	–	S	–	S	S	–	–	–	S	–
Gentamicin	S	S	–	S	–	–	–	–	–	–	–	–	–	–	S	–	S	S	–	–	–	S	–
Nitrofurantoin	S	S	–	S	–	–	–	–	–	–	–	–	–	–	S	–	S	S	–	–	–	S	S
Piperacillin-tazobactam	R	R	–	R	–	–	–	–	–	–	–	–	–	–	R	–	R	R	–	–	–	S	–
Tobramycin	S	S	–	S	–	–	–	–	–	–	–	–	–	–	S	–	S	S	–	–	–	S	–
ESBL	Class A	Class A	–	Class A	–	–	–	–	–	–	–	–	–	Class A	–	Class A	Class A	–	–	–	–	ND	–

E coli, *Escherichia coli*; GBS, group B Streptococcus; E faecalis, *Enterococcus faecalis*; S haemolyticus, *Staphylococcus haemolyticus*; CFU, colony forming unit; ESBL, extended spectrum beta-lactamase; R, resistant; S, susceptible; NG, no growth; NSG, no significant growth; -, not tested/not available; ND, not detected.

Table 4
Notable mutations identified through whole genome sequencing of pre- and post-treatment isolates.

Gene/Protein	Mutation*	Function	Speculative relevance	Impact	Citations
KpsT (Capsular polysaccharide ABC transporter)	Contig_5, 57347, c.389G>A, p.Arg130His	Involved in capsule synthesis and export	Capsule changes can prevent phage adsorption	Missense mutation may alter function, affecting capsule structure/production	[37]
lutA (Aerobactin siderophore receptor)	Contig_31, 8682, c.125G>T, p.Arg42Leu	TonB-dependent siderophore receptor, potentially hijacked by phages	Alterations could block phage entry while maintaining iron uptake	Missense mutation may disrupt receptor binding sites	[38]
StpA (DNA-binding protein)	Contig_24, 34688, c.340C>T, p.Arg114Cys	Nucleoid organization and stress response regulation, including CRISPR-Cas	Changes may enhance survival under phage stress	Missense mutation could modulate DNA-binding specificity	[39]
PBP1b (Multimodular transpeptidase-transglycosylase)	Contig_7, 160439, c.1489G>A, p.Gly497Ser	Cell wall biosynthesis	Could affect susceptibility to phage-encoded lysins or interactions	Missense mutation may alter enzyme activity or substrate specificity	[40]

* Pre-treatment genotype numbering (BioSample Accession SAMN47439423).

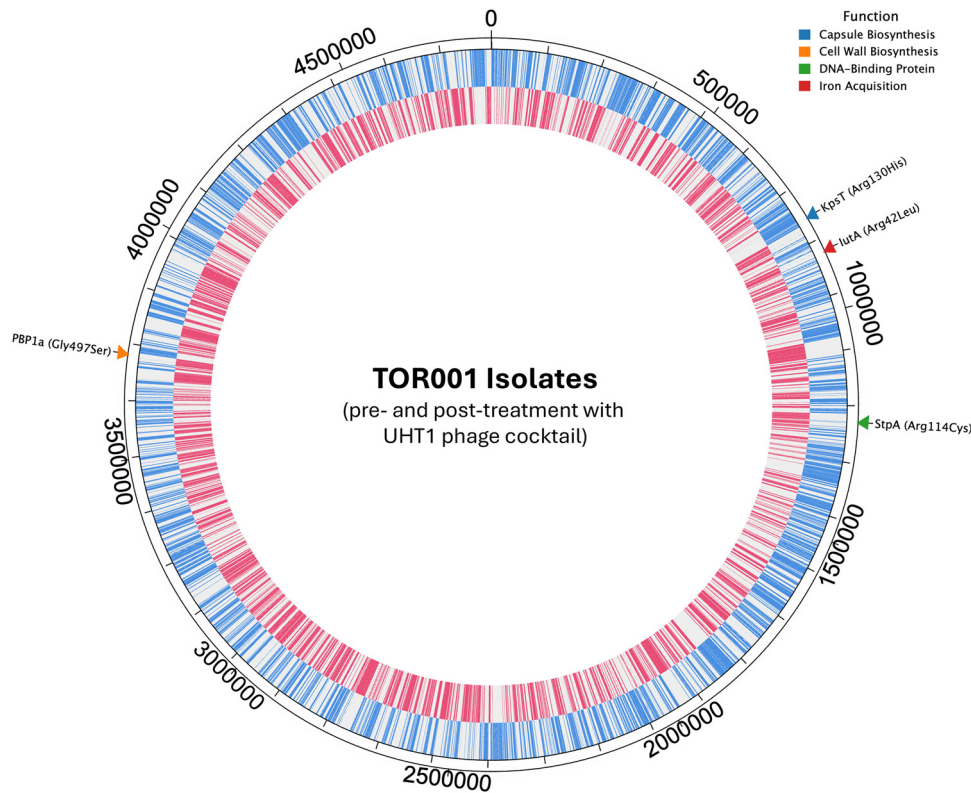


Fig. 3. Circos plot comparing sequencing results for *E. coli* isolates in the TOR001 SPS. The genomic localization of potentially significant mutations identified following treatment with UHT1 phage cocktail. All mutations were missense and are colour-coded by function. The plot was created using a custom script in Python with pyCirclize (1.9.1), matplotlib (v3.9.4), and pandas (v2.3.0).

E. coli. In the TOR001 SPS, treatment with UHT1 phage cocktail met pre-defined safety endpoints. Furthermore, the successive administration of antibiotics following PT resulted in clinical and microbiological cure, the first time in 6 years that the participant had not had a chronic infection with ESBL *E. coli*.

Chronic and recurrent UTIs are thought to occur due to exogenous sources such as sexual intercourse, intracellular bacterial reservoirs in the bladder epithelium, colonization of the genitourinary tract, and colonization of the gastrointestinal tract [32–34]. In the current study design, we administered PT via bladder instillation, oral, and topical application, ostensibly targeting bladder reservoirs, gastrointestinal colonization, and genitourinary colonization, respectively. Decolonization of the gastrointestinal tract using oral PT is supported by pre-clinical work in humanized mice. In this model, 3 daily doses of targeted oral PT had an approximately 50% long-term efficacy in the clearance of multi-drug resistant Enterobacterales [35]. Further, computational modeling of

topical PT applied at the urethral opening is predicted to deliver PT to the urethra and bladder and clear bacterial reservoirs within 36 hours of application [36]. Instilled phage therapy delivers active phage directly to the site of infection within the urinary bladder and is less likely to induce a host neutralizing immune response when compared with intravenous administration [19]. A recent dose-finding study for engineered PT in UTI identified a once daily dosing regimen over two days for bladder instillation of PT [30], however this was specific to their study and did not include a single bladder instilled dose arm in their pharmacokinetics evaluation. Taken together, these preclinical and clinical data support multiple routes for administration of PT at dosing regimens used in the current study.

In practice, PT for UTI treatment has been shown to be safe in human clinical trials and case reports [31]. Our safety and tolerability results are in keeping with two other recent studies where PT was administered via bladder instillation [6,30], although nei-

ther of these studies administered PT orally or via topical application. The most common AEs reported by the patient were pruritic rashes on the bilateral hands and painful rashes on the genitalia. The rashes on the hands were reported 8 and 18 days following initial topical exposure to the UHT1 phage preparation and readily resolved with clotrimazole/betamethasone cream. This AE was considered to have a possible relationship to phage treatment given the indirect exposure and loose relationship with temporality. Irritant contact dermatitis would not be expected to affect the dorsal aspects of the hands symmetrically nor was there any repeat exposures prior to recrudescence of rash on study day 19. The possibility of an allergic (type IV) contact dermatitis cannot be excluded. The genital manifestations occurred approximately 6 and 9 weeks after the last topical application of PT with a duration of approximately 3 days on each occurrence. Given a history of atrophic vaginitis and because UHT1 was administered topically on study days 2, 3, and 4, the DSMB considered these AEs unlikely to be related to the study material. Additionally, the patient reported symptoms of mild diverticulitis that was managed entirely as an outpatient by her family practitioner. When this AE was brought to the attention of the DSMB and discussed, it was considered unlikely to be related to the study material based on temporality and her known history of diverticulosis/diverticulitis. Of note, the patient had decreased eGFR at baseline, which normalized over the course of the study. We cannot solely attribute this to the clearance of her UTI as her daily fluid intake was increased during treatment for diverticulitis. The SPS framework used herein provided significant oversight in study design and facilitated rigorous safety assessments within the scope of the study.

There have been mixed reports of the efficacy of PT when administered via bladder instillation. The largest of these studies was a randomized controlled trial comparing the efficacy of a commercial preparation of phage for twice-daily intravesicular administration in male patients with complex UTI following transurethral resection of the prostate (TURP). This trial showed non-inferiority to antibiotic standard of care but did not outperform placebo containing preservative, which was attributed to inadequate concentrations of active phage in the preparation and therapeutic benefits derived from the placebo treatment [6]. In contrast, the most recent results from the ELIMINATE open-label phase II trial showed high rates of clinical and microbiological cure in all participants who received personalized phage via bladder instillation with concomitant trimethoprim-sulfamethoxazole antibiotics [30]. Co-administration of antibiotics with PT has been demonstrated to be significantly more effective at microbiological cure in a recent case series of 100 patients from multiple centers [3]. A 4-phage cocktail similar to UHT1 was intravenously administered concomitantly with a 6-week course of ertapenem to treat a chronic UTI caused by ESBL *E. coli* in a solid organ transplant recipient to good effect [19]. Taken together, personalized PT in conjunction with antibiotics can lead to microbiological eradication and clinical cure, as was observed in our study.

Our WGS results are similar to those observed in a report describing treatment with intravenous PT and concomitant ertapenem, where clinical cure, but not microbiologic eradication, was observed. Symptom resolution was accompanied by mutations in the causative organism within adhesion and invasion genes, ostensibly reducing virulence and promoting clinical resolution [19]. Phenotypic phage susceptibility results showed the emergence of resistance to HP3.1 between pre- and post-treatment isolates without concomitant mutations in *OmpA*, which had been seen previously in HP3.1 resister [28]. The impact of these mutations is speculative; however, the post-treatment isolate was eradicated following a 2-week course of ertapenem which had not been possible prior to treatment with the UHT1 cocktail.

5. Limitations

The TOR001 SPS is by definition a study in a single patient under the framework provided by Canadian regulatory bodies and is not intended to provide statistically robust data or to be used as a basis for general clinical decisions.

Exploratory analysis of *E. coli* isolates was limited by technical issues at our clinical laboratories. Specifically, stool specimens were not assessed for the presence of ESBL *E. coli*, making it impossible to assess the efficacy of oral administration of the UHT1 cocktail on ESBL *E. coli* carriage in the gut. Optimization of healthy vaginal flora is a mainstay of chronic UTI treatment. Although our study assessed vaginal swabs using microscopy, genital cultures to identify the presence of ESBL *E. coli* within the vaginal flora were not performed. Colonization of the gastrointestinal tract and genitourinary tract are known risk factors for recurrent urinary tract infections, however we cannot state if the administration of PT in the TOR001 SPS created changes in ESBL *E. coli* carriage.

6. Conclusions

PT holds great promise as a targeted and adaptable strategy against resistant and indolent infections such as ESBL *E. coli*. While challenges exist, advancements in understanding host-phage interactions and production techniques will propel PT into a viable and potent tool in the battle against antimicrobial resistance. The synergy between PT and existing interventions, along with careful consideration of safety and ease of implementation, will be pivotal in realizing the full potential of PT in multi-drug resistant infections.

Funding

This work was supported by the Laboratory Associates Fund of the Unity Health Toronto, the [Canadian Institutes of Health Research](#) (Pandemics and Health Emergencies Grant number 520161), and the professorship in Bacteriophage Therapy Research and Innovation, University of Toronto. LLB holds a [Canada Research Chair](#) in Microbe-Surface Interactions (CRC 2021-00103), KLM holds a [Canada Research Chair](#) in Bacteriophage Biology and Therapeutics (CRC 2023-00010). ARD is the [Canada Research Chair](#) in Bacteriophage-Based Technologies (950-232058).

Ethical approval: Unity Health Toronto Research Ethics Board - Approval received 21-Feb-2023; REB 22-210.

Sequence information: Whole genome sequences for pre- and post-treatment *E. coli* isolates have been deposited in the NCBI GenBank: TOR001 (BioSample: SAMN47439423, GenBank accession: JBMOKN000000000); TOR001.1 (BioSample: SAMN47439424, GenBank accession: JBMOKM000000000).

In memoriam: The authors dedicated this work to the memory of Dr. Parviz Mohammed Sabour (1936 – 2025) a phage scientist and an ardent supporter of phage applications globally.

Declaration of competing interests: Declaration of Interest for authors JDC, PBH, KCS, JRC, UJ, CF, HH, UB, AMK, ARD, KLM, LLB, ALT, and AWM: none

The corresponding author, GJG, is the co-founder of the non-profit Phage Canada and serves as an elected officer on the board. GJG is a co-founder and chair of Global Clinical Phage Rounds a group of more than 400 professionals who discuss cases related to phage therapy and access. GJG is the principal shareholder and president for medical corporations in his name located in Ontario,

and Prince Edward Island Canada. These corporations provide clinical and laboratory services on urinary tract infections and other infections.

Acknowledgments: The authors would like to thank the Unity Health Toronto laboratory, pharmacy, and nursing clinical staff as well as Bay Area Research Logistics and Health Canada. We acknowledge Dr. Mark A. Downing, Dr. Kevin Schwartz, Dr. Melanie Tsang, and Dr. Travis Carpenter from St. Joseph's Health Centre who participated in the DSMB. We recognize Jenna Wong for her role as the Association of Medical Microbiology and Infectious Disease Canada Clinical Research Network Research Coordinator. The following laboratory and research laboratory staff at Unity Health Toronto provided support for the study: Dr. Catherine Streutker, Dr. Larissa Matukas, Matthew Doggart, Catherine Park, Tijan Jallow, Mark Bignell, and Jenny Wu. We would also acknowledge the pharmacy at Unity Health Toronto for their assistance, in particular Clarence Chant, Jiten Jain, and Glenda Bennett. Gurpreet Lakhanpal at Applied Health Research Centre (Unity Health Toronto) provided study support. The authors would like to thank Dr. Stefanie Strathdee and Dr. Saima Aslam from the Centre for Innovative Phage Applications and Therapeutics, Univ. of California, San Diego for critical advice. We acknowledge the Biologic and Radiopharmaceutical Drug Directorate, Health Canada for their regulatory support. We are grateful to Patryk Aftanas and Dr. Rob Kozak at Shared Hospital Laboratories for technical assistance with bacterial whole genome sequencing. Finally, we would like to thank the participant and her family.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.ijantimicag.2025.107613](https://doi.org/10.1016/j.ijantimicag.2025.107613).

References

- [1] Uytendaele S, Chen B, Onseal J, Ruthooren F, Debaveye Y, Devolder D, et al. Safety and efficacy of phage therapy in difficult-to-treat infections: a systematic review. *Lancet Infect Dis* 2022;22(8):e208–20.
- [2] Broncano-Lavado A, Santamaría-Corral G, Esteban J, García-Quintanilla M. Advances in bacteriophage therapy against relevant MultiDrug-resistant pathogens. *Antibiotics (Basel)* 2021;10(6):672.
- [3] Pirnay JP, Djebbara S, Steurs G, Griselain J, Cochez C, De Soir S, et al. Personalized bacteriophage therapy outcomes for 100 consecutive cases: a multicentre, multinational, retrospective observational study. *Nat Microbiol* 2024;9(6):1434–53.
- [4] Chan BK, Turner PE, Kim S, Mojibian HR, Elefteriades JA, Narayan D. Phage treatment of an aortic graft infected with *Pseudomonas aeruginosa*. *Evol Med Public Health* 2018;2018(1):60–6.
- [5] Petrovic Fabijan A, Lin RCY, Ho J, Maddocks S, Ben Zakour NL, Iredell JR. Safety of bacteriophage therapy in severe *Staphylococcus aureus* infection. *Nat Microbiol* 2020;5(3):465–72.
- [6] Leitner L, Ujmajuridze A, Chanishvili N, Goderdzishvili M, Chkonia I, Rigvava S, et al. Intravesical bacteriophages for treating urinary tract infections in patients undergoing transurethral resection of the prostate: a randomised, placebo-controlled, double-blind clinical trial. *Lancet Infect Dis* 2021;21(3):427–436.
- [7] Chan BK, Stanley GL, Kortright KE, Vill AC, Modak M, Ott IM, et al. Personalized inhaled bacteriophage therapy for treatment of multidrug-resistant *Pseudomonas aeruginosa* in cystic fibrosis. *Nat Med* 2025;31(5):1494–501.
- [8] Pires DP, Melo L, Vilas Boas D, Sillankorva S, Azeredo J. Phage therapy as an alternative or complementary strategy to prevent and control biofilm-related infections. *Current Opinion in Microbiology* 2017;39:48–56.
- [9] Gu Liu C, Green SI, Min L, Clark JR, Salazar KC, Terwilliger AL, et al. Phage-antibiotic synergy is driven by a unique combination of antibacterial mechanism of action and stoichiometry. *mBio* 2020;11(4).
- [10] El Ghali A, Stamper K, Kunz Coyne AJ, Holger D, Kebriaei R, Alexander J, et al. Ciprofloxacin in combination with bacteriophage cocktails against multi-drug resistant *Pseudomonas aeruginosa* in ex vivo simulated endocardial vegetation models. *Antimicrob Agents Chemother* 2023;67(11):e0072823.
- [11] Leslie M, Fadaak R, Lethebe BC, Szostakiwskyj JH. Assessing the appropriateness of community-based antibiotic prescribing in Alberta, Canada, 2017–2020, using ICD-9-CM codes: a cross-sectional study. *CMAJ Open* 2023;11(4):E579–86.
- [12] Leis JA, Born KB, Ostrow O, Moser A, Grill A. Prescriber-led practice changes that can bolster antimicrobial stewardship in community health care settings. *Can Commun Dis Rep* 2020;46(1):1–5.
- [13] Low M, Neuberger A, Hooton TM, Green MS, Raz R, Balicer RD, et al. Association between urinary community-acquired fluoroquinolone-resistant *Escherichia coli* and neighbourhood antibiotic consumption: a population-based case-control study. *Lancet Infect Dis* 2019;19(4):419–28.
- [14] German GJ, Frenette C, Caissy JA, Grant J, Lefebvre MA, Mertz D, et al. The 2018 global point prevalence Survey of antimicrobial consumption and resistance in 47 Canadian hospitals: a cross-sectional survey. *CMAJ Open* 2021;9(4):E1242–51.
- [15] Dason S, Dason JT, Kapoor A. Guidelines for the diagnosis and management of recurrent urinary tract infection in women. *Can Urol Assoc J* 2011;5(5):316–22.
- [16] Mysorekar IU, Hultgren SJ. Mechanisms of uropathogenic *Escherichia coli* persistence and eradication from the urinary tract. *Proc Natl Acad Sci* 2006;103(38):14170–5.
- [17] Sanchez CJ Jr, Mende K, Beckius ML, Akers KS, Romano DR, Wenke JC, et al. Biofilm formation by clinical isolates and the implications in chronic infections. *BMC Infect Dis* 2013;13:47.
- [18] Lutgring JD. Carbapenem-resistant Enterobacteriaceae: An emerging bacterial threat. *Semin Diagn Pathol* 2019;36(3):182–6.
- [19] Terwilliger A, Clark JR, Karris M, Hernandez-Santos H, Green S, Aslam S, et al. Phage therapy related microbial succession associated with successful clinical outcome for a recurrent urinary tract infection. *Viruses* 2021;13(10):2049.
- [20] Aziz RK, Bartels D, Best AA, DeJongh M, Disz T, Edwards RA, et al. The RAST Server: rapid annotations using subsystems technology. *BMC Genomics* 2008;9:75.
- [21] Overbeek R, Olson R, Pusch GD, Olsen GJ, Davis JJ, Disz T, et al. The SEED and the rapid annotation of microbial genomes using subsystems technology (RAST). *Nucleic Acids Res* 2014;42(Database issue):D206–14.
- [22] Brettin T, Davis JJ, Disz T, Edwards RA, Gerdes S, Olsen GJ, et al. RASTtk: a modular and extensible implementation of the RAST algorithm for building custom annotation pipelines and annotating batches of genomes. *Sci Rep* 2015;5:8365.
- [23] Jolley KA, Bray JE, Maiden MCJ. Open-access bacterial population genomics: BIGSdb software, the PubMLST.org website and their applications. *Wellcome Open Res* 2018;3:124.
- [24] Roer L, Tchesnokova V, Allesøe R, Muradova M, Chattopadhyay S, Ahrenfeldt J, et al. Development of a Web Tool for *Escherichia coli* Subtyping Based on. *J Clin Microbiol* 2017;55(8):2538–43.
- [25] Katoh K, Standley DM. MAFFT multiple sequence alignment software version 7: improvements in performance and usability. *Mol Biol Evol* 2013;30(4):772–780.
- [26] Darling AC, Mau B, Blattner FR, Perna NT. Mauve: multiple alignment of conserved genomic sequence with rearrangements. *Genome Res* 2004;14(7):1394–403.
- [27] Waddell TE, Franklin K, Mazzocco A, Johnson RP. Preparation and characterization of anti-phage serum. In: Clokie MJ, Kropinski AM, editors. *Bacteriophages Methods and Protocols*, Volume 1: Isolation, Characterization, and Interactions. New York, NY: Humana Press; 2009. p. 287–92.
- [28] Salazar KC, Ma L, Green SI, Zulk JJ, Trautner BW, Ramig RF, et al. Antiviral resistance and phage counter adaptation to antibiotic-resistant extraintestinal pathogenic. *mBio* 2021;12(2):e00211–21.
- [29] Sekizuka T, Tanaka R, Hashino M, Yatsu K, Kuroda M. Comprehensive genome and plasmidome analysis of antimicrobial resistant bacteria in wastewater treatment plant effluent of Tokyo. *Antibiotics (Basel)* 2022;11(10):1283.
- [30] Kim P, Sanchez AM, Penke TJR, Tuson HH, Kime JC, McKee RW, et al. Safety, pharmacokinetics, and pharmacodynamics of LBP-EC01, a CRISPR-Cas3-enhanced bacteriophage cocktail, in uncomplicated urinary tract infections due to *Escherichia coli* (ELIMINATE): the randomised, open-label, first part of a two-part phase 2 trial. *Lancet Infect Dis* 2024;24(12):1319–1332.
- [31] Al-Anany AM, Hooley PB, Cook JD, Burrows LL, Martyniuk J, Hynes AP, et al. Phage therapy in the management of urinary tract infections: a comprehensive systematic review. *Phage (New Rochelle)* 2023;4(3):112–27.
- [32] Brannon JR, Dunigan TL, Beebout CJ, Ross T, Wiebe MA, Reynolds WS, et al. Invasion of vaginal epithelial cells by uropathogenic *Escherichia coli*. *Nat Commun* 2020;11(1):2803.
- [33] Glover M, Moreira CG, Sperandio V, Zimmern P. Recurrent urinary tract infections in healthy and nonpregnant women. *Urol Sci* 2014;25(1):1–8.
- [34] Stapleton AE. The vaginal microbiota and urinary tract infection. *Microbiol Spectr* 2016;4(6).
- [35] Liu JY, Lin TL, Chiu CY, Hsieh PF, Lin YT, Lai LY, et al. Decolonization of carbapenem-resistant *Klebsiella pneumoniae* from the intestinal microbiota of model mice by phages targeting two surface structures. *Front Microbiol* 2022;13:877074.
- [36] Blanco C, Chen IA. Phage therapy administered noninvasively could be effective in thin tubes subject to episodic flow despite washout: a simulation study. *Phys Biol* 2019;16(5):054001.

- [37] Arredondo-Alonso S, Blundell-Hunter G, Fu Z, Gladstone RA, Fillol-Salom A, Lorraine J, et al. Evolutionary and functional history of the *Escherichia coli* K1 capsule. *Nat Commun* 2023;14(1):3294.
- [38] Robinson AE, Heffernan JR, Henderson JP. The iron hand of uropathogenic *Escherichia coli*: the role of transition metal control in virulence. *Future Microbiol* 2018;13(7):745–56.
- [39] Mitić D, Radović M, Markulin D, Ivančić-Baće I. StpA represses CRISPR-Cas immunity in H-NS deficient *Escherichia coli*. *Biochimie* 2020;174:136–43.
- [40] King DT, Wasney GA, Nosella M, Fong A, Strynadka NCJ. Structural Insights into Inhibition of *Escherichia coli* Penicillin-binding Protein 1B. *J Biol Chem* 2017;292(3):979–93.